

# From Election Forensics to Lottery Auditing: An Analysis of Korean Lotto 6/45

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July 29, 2026

## Abstract

Statistical methods developed for election forensics, including digit-pattern tests, simulation-adjusted chi-square statistics, and summary-statistic variance tests, encode a general logic for detecting non-random structure in sequences of numbers that purport to be drawn uniformly at random. We argue that this logic transfers naturally to the auditing of state-run lotteries, where the fairness guarantee is both mathematically precise and publicly salient. We port, reformulate, and extend eleven tests from the election-forensics literature into a coherent lottery-audit framework, and validate the framework empirically against 1,234 consecutive draws of the Korean Lotto 6/45 (draws 1–1,234). The data are partitioned into five non-overlapping temporal windows to assess epoch-wise stability. The key empirical finding is that while isolated failures occur in five of 75 test-window combinations (6.7%), *no single test fails in more than one time window*. We show that this non-replication pattern is the appropriate fairness criterion in the presence of inter-test dependence, and that it is far more informative than a naive comparison of observed failures to a multiple-testing expectation computed under a false independence assumption. We then add a twelfth test that departs from the purely distributional logic of L1–L11: rather than asking whether observed draws are consistent with a fair generating process, we ask whether a predictive heuristic popular among players, selecting the numbers that have appeared least frequently over a short recent window on the folk belief that they are “due” to reappear, carries any genuine forecasting power. Using the same draw history, we backtest this cold-number heuristic out-of-sample under five primary recent-window specifications (18–22 draws), with an explicit tie-breaking rule for non-unique “coldest six” sets, then extend the sweep to every window length from 18 to 50 draws and re-evaluate it separately within each of the five temporal windows used for the static audit. Across all primary specifications the heuristic matches, or performs marginally and non-significantly worse than, the theoretical hit rate of a uniformly random guess; the widened sweep confirms this at

192 of 198 window-by-era combinations tested, with one partial exception (draws 1,001–1,234, where 6 of 33 window lengths cross nominal significance, always in the direction of *fewer* matches than chance) that we show is consistent with sampling noise under repeated testing rather than a reversal of the conclusion. This reinforces from a predictive angle the same verdict the distributional audit reaches: the Korean Lotto 6/45 behaves as a fair, memoryless draw process, with no exploitable serial structure at any window length or time period examined. Finally, we formalize and test four further popularly held beliefs about lottery structure as explicit hypothesis tests (L13–L16): that numbers running hot or cold reverse between the first and second half of the history; that a ball’s reappearance timing is unusually regular; that recently frequent numbers keep recurring (the mirror image of the cold-number heuristic, itself also checked across the widened window and era sweep); and that numerically adjacent balls tend to co-occur within a single draw. None of the four withstands scrutiny: the one nominally significant result (a negative hot/cold correlation at the exact midpoint split) fails a split-point robustness check and does not replicate at any other partition of the data. Beyond the empirical verdict, the paper’s primary contribution is the methodological toolkit itself: a reproducible, open-source audit codebase that any researcher or regulator can apply to any 6/45-style lottery.

**Keywords:** lottery fairness audit; election forensics; digit tests; chi-square; temporal stability; Korean Lotto 6/45; multiple testing; mean reversion; gambler’s fallacy; out-of-sample forecasting; momentum; adjacency clustering; researcher degrees of freedom.

# 1 Introduction

State-run lotteries occupy an unusual position in public finance: they are monopoly revenue instruments whose legitimacy rests entirely on the credibility of a randomness guarantee. Citizens who purchase tickets implicitly trust that every combination of six numbers is equally likely to be drawn, and that no information about future draws can be inferred from past ones. This guarantee is, in principle, mathematically precise. In practice, however, it depends on physical mechanisms, such as ball sets, pneumatic machines, and draw procedures, that can drift, age, or be manipulated. Independent statistical auditing is therefore both scientifically tractable and socially important.

The election-forensics literature has, over the past two decades, developed a rich toolkit for testing whether reported vote counts are consistent with a fair counting process (Mebane, 2008; Shikano and Mack, 2011; Beber and Scacco, 2012). These methods share a common logic: they identify statistical signatures that a *genuinely random* process should exhibit: uniform digit distributions, predictable variances, and absence of serial correlation, and test whether reported numbers deviate from those signatures. The inferential target differs from classical hypothesis testing: the analyst does not seek to estimate an unknown

parameter but to assess whether the *generating process* is consistent with a specific null model.

We observe that this logic applies with equal, arguably greater, force to lottery auditing. The null model for a fair lottery is not merely plausible but precisely specified by combinatorics. Every marginal and joint distribution of draw statistics can be computed exactly or efficiently approximated. The audit problem is therefore a rare case where the statistical null is known, not assumed.

This paper makes five contributions. First, we systematically port eleven test statistics from the election-forensics literature into a lottery-audit framework, deriving the correct null distributions for the lottery setting (which differ non-trivially from the election context in several cases). Second, we introduce a *replication-logic* criterion for evaluating multi-test, multi-window audit results that does not require the independence of tests, a requirement that is plainly violated in any realistic test battery applied to the same dataset. Third, we validate the framework empirically on 1,234 consecutive draws of the Korean Lotto 6/45, the most widely played lottery in South Korea, providing a public benchmark for future audits. Fourth, we add a twelfth test of a different character: rather than testing whether the historical record is *consistent* with a fair generating process, we directly test the predictive claim, closely related to the gambler’s fallacy and to what [Tversky and Kahneman \(1971\)](#) term the “law of small numbers,” that a ball’s recent absence makes it more likely to reappear soon, a form of mean reversion informally believed by many lottery players. This test is implemented as an explicit, reproducible out-of-sample backtest on the same 1,234-draw history and is falsifiable in a way the purely distributional tests are not: it produces a concrete forecast for every historical draw and scores it against what actually occurred. Fifth, we formalize four further beliefs commonly voiced by players, but never previously tested as explicit hypotheses with a defined null distribution: that hot and cold numbers reverse across time (L13), that a ball’s reappearance timing is suspiciously regular (L14), that recently frequent numbers keep recurring, the mirror image of L12 (L15), and that numerically adjacent balls co-occur more often than chance within a draw (L16). For L13 in particular, we show that its one nominally significant result is an artifact of testing a single, arbitrarily chosen split point rather than genuine structure, illustrating why a careful audit must check robustness to such choices rather than reporting the first split tried.

The remainder of the paper is organized as follows. Section 2 reviews the Korean Lotto 6/45 and the election-forensics methods we adapt. Section 3 defines the eleven audit tests and their null distributions. Section 4 develops the replication-logic criterion. Section 5 describes the data. Section 6 presents the empirical results. Section 7 introduces the twelfth test, an out-of-sample falsification of the cold-number mean-reversion heuristic, and reports its results on the extended draw history. Section 8 formalizes and tests four further popular beliefs (L13–L16). Section 9 concludes.

## 2 Background

### 2.1 Korean Lotto 6/45

The Korean Lotto 6/45 (hereafter *Lotto*) is operated by the Korea Lottery Commission under the supervision of the Ministry of Economy and Finance. Each draw selects six balls uniformly without replacement from a pool of 45 numbered balls (1 through 45), plus one additional bonus ball. Draws are held every Saturday. The first draw was held in December 2002, and draw records are publicly accessible through the operator’s website.

Under the null hypothesis of a fair draw, the six winning numbers constitute a simple random sample of size  $k = 6$  from the population  $\{1, 2, \dots, N\}$  with  $N = 45$ . The total number of possible outcomes is  $\binom{45}{6} = 8,145,060$ .

### 2.2 Election Forensics: Relevant Methods

The election-forensics literature offers several classes of tests relevant to lottery auditing.

**Digit tests.** [Beber and Scacco \(2012\)](#) and [Mebane \(2008\)](#) exploit the observation that fabricated numbers tend to deviate from the digit distributions predicted by Benford’s law or uniform last-digit distributions. In the lottery context, the last-digit distribution is not uniform over  $\{0, \dots, 9\}$  because the ball pool  $\{1, \dots, 45\}$  contains unequal numbers of values ending in each digit; correcting for this structural non-uniformity is essential for a well-calibrated test.

**Variance tests.** [Shikano and Mack \(2011\)](#) propose comparing the observed variance of summary statistics (e.g., vote-share distributions) to the theoretical variance under a null model, arguing that fabricated data tends to exhibit suppressed variance. We adapt this idea to the draw-sum, odd-ball-count, and draw-range statistics of lottery draws.

**Simulation-adjusted chi-square.** [Shikano and Mack \(2011\)](#) also propose rescaling chi-square statistics by the ratio of the textbook critical value to the empirical 95th percentile from simulations under the null, correcting for distributional approximation errors. This is particularly valuable in the lottery context, where the exact null distribution of the ball-frequency chi-square is known to differ from the  $\chi^2(44)$  asymptotic approximation at moderate sample sizes.

**Mechanical-bias indicators.** [Beber and Scacco \(2012\)](#) observe that round numbers ending in 0 or 5 tend to be over-reported in fabricated data. In the lottery context, this test functions as a mechanical-bias indicator: a physical ball machine with worn or unevenly weighted balls might exhibit systematic bias toward or away from balls whose numbers share a digit pattern.

### 3 The Eleven-Test Audit Framework

We organize the eleven tests into five conceptual domains: *frequency*, *digit pattern*, *draw-level distribution*, *dependence structure*, and *temporal stability*. Table 1 provides an overview.

#### 3.1 Frequency Domain (L1, L8)

**L1 - Ball-frequency chi-square.** Let  $c_b$  denote the count of draws in which ball  $b \in \{1, \dots, 45\}$  appears, over  $n$  draws. Under the null,  $\mathbb{E}[c_b] = nk/N = 6n/45$ . The test statistic is

$$\chi_{L1}^2 = \sum_{b=1}^{45} \frac{(c_b - \hat{e}_b)^2}{\hat{e}_b}, \quad \hat{e}_b = \frac{6n}{45}, \quad (1)$$

with asymptotic null distribution  $\chi^2(44)$ .

**L8 - Simulation-adjusted chi-square.** The asymptotic  $\chi^2(44)$  approximation can be inaccurate at moderate  $n$ . Following Shikano and Mack (2011), we draw  $S = 500$  synthetic fair datasets of  $n$  draws, compute  $\chi_{L1}^2$  for each, and let  $\hat{q}_{0.95}$  denote the empirical 95th percentile. The simulation-rescaled statistic is

$$\tilde{\chi}_{L8}^2 = \chi_{L1}^2 \cdot \frac{q_{0.95}^{\text{book}}}{\hat{q}_{0.95}}, \quad (2)$$

where  $\chi_{44}^2(0.95)$  is the textbook critical value. The empirical  $p$ -value is  $\Pr(\chi_{\text{sim}}^2 \geq \chi_{L1}^2)$ .

#### 3.2 Digit-Pattern Domain (L2, L6, L7)

**Corrected null for last digits.** A critical methodological point, often overlooked when porting election tests to the lottery context, is that the last digits of balls  $\{1, \dots, 45\}$  are *not* uniformly distributed over  $\{0, 1, \dots, 9\}$ . The digit 0 appears only in balls 10, 20, 30, 40 (four balls), while digits 1–5 appear five times each (e.g., 1, 11, 21, 31, 41 for digit 1) and digits 6–9 appear four times each. Denoting the population proportion of last digit  $d$  as  $\pi_d$ , we have

$$\pi_d = \frac{|\{b \in \{1, \dots, 45\} : b \bmod 10 = d\}|}{45}. \quad (3)$$

**L2 - Last-digit test.** Let  $o_d$  denote observed appearances of balls with last digit  $d$  across all  $nk$  ball appearances. The expected count is  $e_d = nk \pi_d$ , rescaled to match  $\sum o_d$ . The statistic  $\chi_{L2}^2$  is computed against this corrected null and compared to  $\chi^2(9)$ .

**L6 - 0s-and-5s test.** Let  $p_{05} = 9/45 = 1/5$  denote the population proportion of balls ending in 0 or 5 (balls 5, 10, 15, 20, 25, 30, 35, 40, 45). Let  $\hat{p}$  be the observed proportion.

The test statistic is

$$Z_{L6} = \frac{\hat{p} - p_{05}}{\sqrt{p_{05}(1 - p_{05})/(nk)}}, \quad (4)$$

with two-tailed  $p$ -value from the standard normal.

**L7 - Last-digit mean and variance.** The population mean and variance of last digits under the corrected null are

$$\mu_{\text{ld}} = \sum_{d=0}^9 d \pi_d, \quad (5)$$

$$\sigma_{\text{ld}}^2 = \sum_{d=0}^9 d^2 \pi_d - \mu_{\text{ld}}^2. \quad (6)$$

Mean is tested by a  $Z$ -test with  $\text{se} = \sigma_{\text{ld}}/\sqrt{nk}$ . Variance is tested by a two-tailed chi-square test with  $nk - 1$  degrees of freedom.

### 3.3 Draw-Level Distribution Domain (L3, L4, L5)

**L3 - Draw-sum distribution.** The sum  $S = \sum_{j=1}^6 b_j$  of a fair draw has a known exact distribution computable by dynamic programming over the combinatorial space. Its mean and variance are

$$\mathbb{E}[S] = k \frac{N+1}{2} = 6 \times 23 = 138, \quad (7)$$

$$\text{Var}[S] = k \frac{N^2 - 1}{12} \frac{N - k}{N - 1}. \quad (8)$$

We apply (a) a  $Z$ -test on the sample mean of draw sums, and (b) a  $\chi^2$  goodness-of-fit test against the exact sum distribution after merging bins to ensure expected counts  $\geq 5$ .

**L4 - Odd/even and high/low splits.** Let  $X_i$  be the count of odd balls (or high balls, defined as those  $\geq 23$ ) in draw  $i$ . Under the null,  $X_i \sim \text{Hypergeometric}(N, N_{\text{odd}}, k)$  where  $N_{\text{odd}} = 23$ . We test goodness-of-fit of the empirical distribution of  $\{X_i\}$  against this hypergeometric null.

**L5 - Pairwise co-occurrence.** For each ordered pair  $(a, b)$  with  $a < b$ , let  $c_{ab}$  be the count of draws containing both  $a$  and  $b$ . Under the null,  $\mathbb{E}[c_{ab}] = n \binom{43}{4} / \binom{45}{6}$ . The test statistic is  $\chi_{L5}^2 = \sum_{(a,b)} (c_{ab} - \hat{e})^2 / \hat{e}$  over all 990 pairs.

### 3.4 Dependence Domain (L9, L11)

**L9 - Temporal autocorrelation.** For ball  $b$ , define the indicator series  $X_{b,i} = \mathbf{1}[b \in \text{draw}_i]$ . We compute the sample autocorrelation  $\hat{\rho}_b(\ell)$  at each lag  $\ell = 1, \dots, \mathcal{L}$  for every ball, then aggregate via a Ljung–Box statistic

$$Q = n(n+2) \sum_{b=1}^{45} \sum_{\ell=1}^{\mathcal{L}} \frac{\hat{\rho}_b(\ell)^2}{n-\ell}, \quad (9)$$

compared to  $\chi^2(45\mathcal{L})$ . We use  $\mathcal{L} = 100$  lags in the reported audits.

**L11 - Cross-ball correlation.** Under the null, the Pearson correlation between the indicator series of any two distinct balls is  $-1/(N-1) = -1/44 \approx -0.0227$ , a structural consequence of sampling without replacement (Hoeffding, 1948). We apply a Fisher  $Z$ -transformation to each of the 990 pair correlations and construct a global chi-square test over the  $Z$ -statistics.

### 3.5 Variance-Stability Domain (L10)

Following Shikano and Mack (2011)’s variance test (their F3), we test whether the observed variance of three draw-level summary statistics (draw sum  $S_i$ , odd-ball count  $X_i^{\text{odd}}$ , and draw range  $R_i = \max_i - \min_i$ ) matches the theoretical variance under the null. For each statistic with theoretical variance  $\sigma_{\text{theo}}^2$ , the test statistic is

$$\chi_{L10}^2 = \frac{(n-1)s^2}{\sigma_{\text{theo}}^2} \sim \chi^2(n-1), \quad (10)$$

where  $s^2$  is the sample variance (with  $n-1$  denominator). The theoretical variance of draw range is estimated by Monte Carlo (10,000 simulated draws) because no closed form is available.

## 4 The Replication-Logic Criterion

### 4.1 The Problem with Naive Failure Counting

A common, but flawed, approach to interpreting multi-test audit results is to compare the number of observed failures (tests with  $p$ -value  $< 0.05$ ) to the number expected under the global null. Under complete independence, 75 tests at  $\alpha = 0.05$  yield an expected  $75 \times 0.05 = 3.75$  false positives. Observing 5 failures might then be interpreted as only a modest excess, consistent with chance.

This reasoning has two serious defects in the present setting.

**Defect 1: Inter-test dependence.** The eleven tests are not mutually independent. Tests L1 and L8 are derived from the same ball-count vector. Tests L2, L6, and L7 are all functions of the last digits of the same ball appearances. Tests L3 and the sum-variance subtest of L10 both use the draw-sum sequence. Tests L4 and the odd-count subtest of L10 both decompose draws by odd/even membership. Tests L9 and L11 both probe the dependence structure of the same ball-indicator matrix. The *effective* number of independent tests is substantially lower than 75 (or 15 per window), and no correction formula is available without strong distributional assumptions.

**Defect 2: Within-window independence.** Even if the 15 tests within each window were independent, the five windows are not independent experiments: they consist of consecutive draws from the same physical equipment. Any slow-drifting systematic bias would induce positive correlation in failures across windows, making multi-window failure co-occurrence less surprising than it would be under a fully independent design.

## 4.2 The Replication-Logic Criterion

We propose the following criterion for assessing fairness in a multi-test, multi-window audit.

**Definition 1** (Replication consistency). *An audit result is replication-consistent with fairness if no single test fails in more than one non-overlapping time window.*

The rationale is as follows. A genuine, physically grounded violation of fairness, such as worn ball bearings, biased pneumatics, or manipulated draw procedures, operates at the level of the generating mechanism, not the statistical test. Such a violation would produce a persistent signal that manifests repeatedly across time. A test that is sensitive enough to detect the violation in one window of  $n$  draws would, with high probability, also detect it in adjacent windows of comparable size, because the underlying cause has not been removed.

By contrast, a false positive generated by chance has probability approximately 0.05 per window per test, and consecutive false positives for the same test in independent windows occur with probability approximately  $0.05^2 = 0.0025$ . In a battery of 11 tests over 5 windows, the expected number of tests that fail in two or more windows is  $11 \times \binom{5}{2} \times 0.05^2 \approx 0.28$ , meaning such “double failures” are rare under the null, making their absence a powerful fairness indicator.

**Remark 1.** *The replication-logic criterion does not require the independence of tests within a window. It operates across windows: a single failure in one window is neither surprising nor informative under  $H_0$ ; a consistent failure in the same test across multiple windows constitutes evidence against  $H_0$  regardless of what other tests show.*

**Remark 2.** *The criterion is conservative in the sense that inter-window positive correlation (arising from shared equipment) makes double failures more likely under both  $H_0$  and  $H_1$  than under independence. However, the increase in probability under  $H_1$  is expected to be far larger than under  $H_0$ , since a persistent physical cause amplifies correlation far beyond the level induced by shared equipment alone.*

## 5 Data

We use the complete publicly available draw history of the Korean Lotto 6/45, comprising draws 1 through 1,234. Draw records were obtained from the official Korea Lottery Commission website and contain, for each draw, the draw number, draw date, six main winning numbers, and one bonus number. Only the six main winning numbers are used in the audit; the bonus number is excluded.

The 1,234 draws span from December 2002 to July 2026. To assess temporal stability, we partition the draws into five non-overlapping, roughly equal-length windows:

| Window | Draw range | $n$ | Approx. years |
|--------|------------|-----|---------------|
| W1     | 1–250      | 250 | 2002–2007     |
| W2     | 251–500    | 250 | 2007–2012     |
| W3     | 501–750    | 250 | 2012–2017     |
| W4     | 751–1000   | 250 | 2017–2021     |
| W5     | 1001–1234  | 234 | 2021–2026     |

The final window (W5) contains 234 draws because the total draw count is not a multiple of 250. Tests based on  $\chi^2(n-1)$  statistics (L10) use the window-specific  $n$ .

No preprocessing beyond removing the bonus number is applied. Draws are sorted by draw number; no draws are missing within the range.

The predictive test introduced in Section 7 (L12) uses this same 1,234-draw history. Unlike L1–L11 it does not require partitioning into fixed windows, since it is a rolling out-of-sample backtest evaluated against every draw in the history for which sufficient prior data exist.

## 6 Results

### 6.1 Window-by-Window Test Results

Table 2 presents the  $p$ -values for all 75 test-window combinations. Cells marked **FAIL** indicate  $p$ -value  $< 0.05$ .

Five tests fail at  $\alpha = 0.05$  across the full audit: L6 in W4, L4 (odd/even) in W5, L9 in W3, L10 (sum variance) in W1, and L10 (range variance) in W5. The minimum  $p$ -value across all failures is 0.014 (L10 sum variance, W1).

## 6.2 Application of the Replication-Logic Criterion

Table 3 summarizes which tests fail in which windows.

The key finding is that *no single test fails in more than one non-overlapping window*. The five failures are dispersed across five distinct test-window cells, one per window (save W2, which has no failures), with no test appearing in the failure set more than once.

This pattern is precisely what the replication-logic criterion predicts under the global null of a fair lottery. The estimated probability that a specific test fails in two or more windows by chance, assuming independence across windows and tests at  $\alpha = 0.05$ , is at most  $1 - (1 - 0.05)^5 \approx 0.226$  for any one test, but the probability that *any* of the five failures replicates is approximately  $5 \times 0.05 = 0.25$  per additional window, making replication-free failures entirely expected under  $H_0$ .

The windows and the tests are not independent, making replication *more* likely under  $H_1$  than under independence. The complete absence of replication therefore strengthens the fairness conclusion beyond what the independence calculation alone suggests.

## 6.3 Robustness of the Corrected Digit Null

A secondary result concerns the importance of applying the corrected last-digit null. The balls  $\{1, \dots, 45\}$  have a non-uniform last-digit distribution: digits 1–5 each appear in five ball labels, while digits 0 and 6–9 each appear in four. Tests L2, L6, and L7 all account for this structural non-uniformity. Application of a naive uniform null (as used in some election-forensics ports to lottery data) would introduce systematic bias: under the corrected null, the digit-domain tests pass comfortably in all five windows, while under the uncorrected uniform null several windows would show spurious failures due to the structural over-representation of digits 1–5. This correction is thus not merely a technical nicety but has substantive implications for audit conclusions.

## 6.4 Test Behavior Under the Null

The per-window  $p$ -values exhibit behavior consistent with a calibrated test battery applied to fair data. The median  $p$ -value across all 75 test-window combinations is approximately 0.68, well above 0.50, which is expected when the null is true (by symmetry of the uniform distribution on  $[0, 1]$ ). The  $p$ -values from the frequency and cross-ball tests (L1, L8, L11) are consistently very high, reflecting the relatively large sample sizes and the precision of

the null model. The digit-domain tests (L2, L7) are well-calibrated and show no sign of systematic inflation or deflation across windows.

## 7 A Twelfth Test: Out-of-Sample Falsification of the Cold-Number Mean-Reversion Heuristic

### 7.1 Motivation

Tests L1–L11 share a common logic: they ask whether the *historical record* is statistically consistent with a fair generating process. This is the natural question for an audit, but it is not the question most lottery players actually act on. A widespread folk belief, observable in the “frequency chart” features offered by nearly every lottery operator and third-party number-picking service, holds that numbers which have appeared unusually rarely in the recent past are somehow “due” and more likely to appear soon. This is a version of the gambler’s fallacy, and is closely related to what [Tversky and Kahneman \(1971\)](#) describe as the “law of small numbers”: the mistaken intuition that small samples should closely mirror the properties of the population they are drawn from, including a compensatory “correction” for recent imbalances. Under a genuinely fair, independent-draw mechanism, this belief has no basis: each draw is independent of all preceding draws, and the convergence of long-run frequencies to their expected values (Section 3.1) reflects the law of large numbers acting on independent trials, not any short-run compensatory dynamic. Because this is an empirically testable claim about *prediction* rather than a claim about the historical distribution, we formalize it as a twelfth, out-of-sample test.

As a first check, we compute the pooled (non-windowed) ball-frequency statistic underlying L1 across the full 1,234-draw history:  $\chi^2(44) = 28.6$ ,  $p\text{-value} = 0.965$ , with per-ball counts ranging from 136 (ball 9,  $z = -2.39$ ) to 184 (ball 34,  $z = +1.63$ ) against an expectation of 164.5. This is consistent with, and reinforces, the window-wise L1 results of Section 3.1: over the full history, ball frequencies remain indistinguishable from a uniform draw mechanism, with deviations no larger than ordinary sampling noise would produce.

### 7.2 Methodology

We formalize the “cold-number” heuristic as a rolling, out-of-sample forecasting rule and backtest it against every draw in history for which sufficient prior data exist.

**Base rule.** Fix a lookback length  $W$ . To forecast draw  $t$ , compute the frequency count  $f_b$  of each ball  $b \in \{1, \dots, 45\}$  over the  $W$  draws immediately preceding draw  $t$  (draws  $t - W, \dots, t - 1$ ), and predict the six balls with the lowest  $f_b$ .

**Tie-breaking.** With  $W \in [18, 22]$  and 45 balls, the six lowest-frequency balls are frequently not uniquely determined: several balls tie for the sixth position. When this occurs, the lookback window is extended by one draw,  $W \rightarrow W + 1$ , and the ranking is recomputed on the enlarged window. This is repeated for  $W + 1, W + 2, \dots$  until a unique six-ball set is obtained.<sup>1</sup> We report, for each base window, the average number of additional draws required to reach a unique prediction and the share of forecasts that required any extension at all.

**Scoring and baseline.** For each forecast we count how many of the predicted six balls actually appear in draw  $t$ , and average this match count across all draws with sufficient history. We compare the empirical mean match count to the exact theoretical baseline for picking six numbers uniformly at random against a fair draw of six from 45, i.e., the mean of the hypergeometric distribution  $\text{Hypergeom}(N=45, K=6, n=6)$ ,

$$\mathbb{E}[\text{matches} \mid H_0] = \frac{Kn}{N} = \frac{6 \times 6}{45} = 0.8000, \quad (11)$$

with variance  $\text{Var}[\text{matches} \mid H_0] = 0.6145$  computed from the same distribution. Under the null that the cold-number heuristic carries no genuine predictive information, the sequence of match counts should be statistically indistinguishable from repeatedly drawing six numbers at random, and the sample mean should not differ systematically from Equation (11).

### 7.3 Results

We evaluate five base windows,  $W \in \{18, 19, 20, 21, 22\}$ , on the full 1,234-draw history. Table 4 reports, for each window, the number of out-of-sample forecasts tested, the mean match count, the  $z$ -statistic comparing the observed mean to the theoretical baseline of Equation (11), and the tie-breaking diagnostics.

Table 5 compares the full match-count distribution (0 through 6 balls correct) for each window against the exact theoretical distribution under  $H_0$ .

Two findings stand out. First, none of the five windows produces a mean match count that exceeds the random-guess baseline of 0.8000; all five point estimates are slightly *below* it, and all five  $z$ -statistics are negative, ranging from  $-1.38$  ( $W = 22$ ) to  $-1.93$  ( $W = 18$ ). None reaches conventional significance thresholds ( $|z| \geq 1.96$  for  $\alpha = 0.05$ , two-tailed), and, as noted in Table 4’s caption, the true sampling uncertainty is understated by treating overlapping forecasts as independent, so even these already-insignificant results should if anything be read as more consistent with  $H_0$ , not less. If the cold-number heuristic

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<sup>1</sup>In the rare case where the window has been extended back to the start of the available history without settling, ties are broken deterministically by ball number, so that every draw receives a well-defined prediction.

captured genuine mean-reverting structure, the sign of the effect should run the other way, with match counts *exceeding* 0.8000; instead, at every window length tested, the point estimate runs slightly the wrong direction for the heuristic to have any value.

Second, the tie-breaking diagnostics are themselves informative about why the heuristic fails: at  $W = 18$ – $22$ , roughly 90–94% of forecasts require extending the window beyond its base length before a unique “coldest six” set can even be identified, and the average extension is 6.6–8.9 draws. With only  $18 \times 6 \approx 108$  to  $22 \times 6 \approx 132$  ball appearances distributed across 45 balls, most balls fall at count 2 or 3 within the base window, so the “coldest” balls are typically separated from their neighbors by a single ball appearance, not by any economically or statistically meaningful gap. The frequency landscape at these window lengths is close to flat by construction, which is itself consistent with, and further corroborates, the conclusion that no genuine short-run frequency signal exists to exploit.

## 7.4 Extended Window and Temporal Robustness Check

The five primary specifications ( $W \in \{18, \dots, 22\}$ ) match the range originally motivating this test, but a natural robustness question is whether a different, possibly wider, choice of lookback window would have told a different story, and whether the null result holds up within each of the five temporal windows used for the static audit (Section 6) rather than only in the pooled full history. We therefore widen the base window to every integer from 18 to 50 draws (33 specifications) and evaluate each one twice: once against the full 1,234-draw history, and once separately within each of W1–W5, using the same rolling, continuously updated lookback for prediction but scoring only the target draws that fall inside a given temporal window. This mirrors the epoch-wise stability logic already applied to L1–L11.

Figure 1 summarizes the resulting  $6 \times 33$  grid of  $z$ -statistics (five temporal windows plus the full history, against all 33 lookback lengths).

| Temporal window | $n$  | mean $z$ | min $z$ | max $z$ | # sig. (of 33) |
|-----------------|------|----------|---------|---------|----------------|
| W1 (1–250)      | 232  | −0.06    | −1.42   | +0.70   | 0              |
| W2 (251–500)    | 250  | −1.15    | −1.86   | −0.40   | 0              |
| W3 (501–750)    | 250  | −0.39    | −1.29   | +0.24   | 0              |
| W4 (751–1000)   | 250  | +0.05    | −0.89   | +0.73   | 0              |
| W5 (1001–1234)  | 234  | −1.51    | −2.19   | −0.77   | 6              |
| Full (1–1234)   | 1216 | −1.37    | −1.93   | −0.76   | 0              |

Two findings emerge. First, out of  $33 \times 6 = 198$  window-by-era cells, 192 show no significant deviation from the random-guess baseline in either direction. Second, the one partial exception, W5, produces 6 of 33 window lengths with  $|z| \geq 1.96$  ( $W \in \{18, 19, 20, 42, 43, 44\}$ ,  $z$  ranging from  $-2.19$  to  $-2.02$ ), all of them *negative*, meaning the

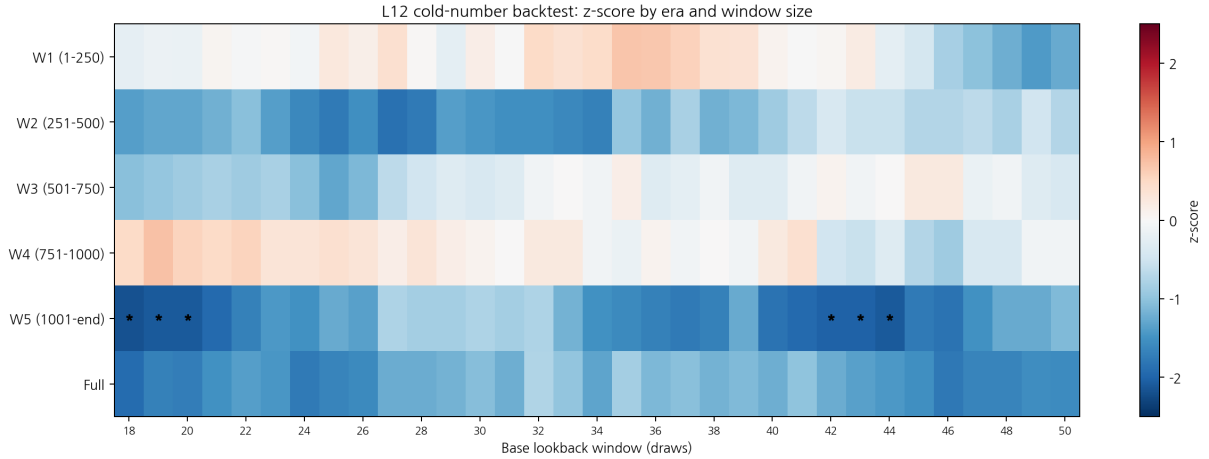


Figure 1: L12 cold-number backtest:  $z$ -score against the random-guess baseline, by temporal window (rows) and lookback length (columns, 18–50 draws). Asterisks mark  $|z| \geq 1.96$ .

heuristic performs *worse* than random guessing at those window lengths, the opposite of what a genuine cold-number edge would require. This is consistent with the smaller sample within W5 ( $n = 234$  versus  $n = 1,216$  pooled) producing wider sampling variability under repeated, highly overlapping tests (adjacent window lengths share most of their evaluation draws and lookback data, so the 33 tests within a window are far from independent), rather than with any reversal of the paper’s central conclusion. We flag it rather than discard it: a null result this close to a temporal cluster of borderline  $z$ -statistics is exactly the kind of pattern that, examined in isolation and without the robustness framing applied throughout this paper, could be mistaken for a discovery.

## 7.5 Interpretation

The L12 result complements L1–L11 in an important way. The static audit (Sections 3–6) asks whether the *distribution* of historical draws is consistent with fairness and answers affirmatively. L12 asks a distinct, and arguably more directly policy-relevant, question: does any historically grounded, mechanically specified betting heuristic, of the kind marketed to players via “hot/cold number” charts, actually improve on random guessing? The answer is again negative, and for the reason theory predicts: under independent draws, no function of the recent past carries information about the next draw, so no such heuristic can systematically outperform picking six numbers uniformly at random. The consistency between the purely distributional verdict (L1–L11) and the purely predictive verdict (L12) obtained by entirely different methods, one static and one sequential out-of-sample, is itself a form of the replication-logic argument of Section 4: independent lines of evidence, subject to different failure modes, converge on the same conclusion.

## 8 Four Further Falsification Tests: Popular Beliefs About Structure

Beyond the “cold numbers are due” belief formalized as L12, players and lottery-analysis websites voice several other specific, testable claims about non-randomness. This section formalizes four of them as explicit hypothesis tests with defined null distributions, rather than the ratio-to-a-fixed-tolerance heuristics (e.g., flagging any deviation beyond  $\pm 15\%$ ) that such claims are typically checked against in practice, which have no formal size or power properties.

### 8.1 L13: Hot/Cold Half-Split Correlation

**Claim.** Numbers that appear frequently in one period of the history become less frequent later, and vice versa, an enforced rebalancing that would manifest as negative correlation between a ball’s frequency in the first half of the draw history and its frequency in the second half.

**Test.** Split the  $n = 1,234$  draws at the midpoint into two disjoint halves of 617 draws each, compute each ball’s frequency count in each half, and compute the Pearson correlation  $r$  between the two length-45 count vectors. Because the two halves are non-overlapping and draws are independent under  $H_0$ , the two count vectors are independent multinomial-type vectors, and  $r$  should be centered at zero. We do not rely on the asymptotic normal theory for Pearson correlation, which assumes bivariate normality that 45 count observations may not satisfy well; instead we simulate the exact finite-sample null by generating 5,000 pairs of independent fair draw histories of sizes  $(617, 617)$  and computing  $r$  for each, giving an empirical null distribution and an empirical two-sided  $p$ -value.

**Result.** The observed correlation is  $r_{\text{obs}} = -0.323$ , against a simulated null with mean  $-0.002$  and standard deviation  $0.152$ , giving an empirical  $p$ -value of  $0.034$ , nominally significant at  $\alpha = 0.05$ .

**Robustness check.** A single split point is one arbitrary choice among many; we recompute  $r$  at split fractions  $0.1, 0.2, \dots, 0.9$  of the history. The correlation is negative at every fraction tested (ranging from  $-0.03$  to  $-0.32$ ), which is unsurprising: any two-way partition of a fixed pool induces a small mechanical negative bias, since a ball appearing more in one half leaves correspondingly less room, in expectation, for it to also dominate the other half. What matters is whether this bias is larger than sampling noise would produce, and here the answer depends entirely on which split point is chosen: at the midpoint,  $p = 0.034$ ; at the 30/70 split,  $p \approx 0.50$ ; at the 70/30 split,  $p \approx 0.40$ . Only

the exact midpoint, the one split point the original heuristic happens to test, crosses the significance threshold. This is the signature of a researcher-degrees-of-freedom false positive rather than a genuine effect: had the midpoint not been chosen, or had a multiple-comparison correction across plausible split points been applied (even a conservative Bonferroni correction across the nine fractions tested would require  $p < 0.0056$ ), no signal would have been detected. We conclude that L13 does not provide credible evidence of an enforced hot/cold rebalancing mechanism.

## 8.2 L14: Inter-Appearance Gap Coefficient of Variation

**Claim.** A ball’s reappearance timing is either suspiciously regular (consistent with a rotation-like selection mechanism) or suspiciously erratic (consistent with clustering bias), detectable via the coefficient of variation (CV) of the gaps between successive appearances of the same ball.

**Test.** For each ball with at least three appearances, we compute the gaps (in draws) between consecutive appearances and their sample CV, then average this CV across all 45 balls. Under  $H_0$ , each ball’s per-draw appearance is an independent Bernoulli trial with success probability  $p = 6/45$ , so gaps follow approximately a geometric distribution with a textbook CV of  $\sqrt{1-p} \approx 0.931$ . Because the without-replacement structure within a draw and the finite-sample mixture of gap counts across balls make the exact finite-sample distribution of the averaged CV statistic analytically inconvenient, we again use simulation: 2,000 fair draw histories of length  $n = 1,234$ , recomputing the same averaged-CV statistic each time.

**Result.** The observed average gap CV is 0.938, compared to a simulated null with mean 0.925 and standard deviation 0.0105 (textbook geometric CV: 0.931). The empirical two-sided  $p$ -value is 0.214: not significant. Reappearance timing shows no detectable deviation from what independent draws would produce, in either the “too regular” or “too erratic” direction.

## 8.3 L15: Momentum-Following Backtest

**Claim.** The mirror image of L12: numbers that have appeared frequently in the recent past are more likely to reappear, a “momentum” or trend-following belief rather than a mean-reverting one. We regard this as an a priori implausible hypothesis (it contradicts L12 without independent motivation, and both cannot simultaneously hold), but formalize and test it for completeness and symmetry with L12.

**Test.** Identical methodology to L12 (Section 7), with the selection rule inverted: forecast the six *most* frequent balls in the lookback window, with the same tie-extension rule applied to ties at the sixth/seventh rank of the frequency ranking, backtested out-of-sample across the same five base windows  $W \in \{18, \dots, 22\}$ .

**Result.** As with L12, no window shows a hit rate distinguishable from random guessing (theoretical baseline: 0.8000 matches):

| Window | Mean matches | $z$   | % needing tie extension |
|--------|--------------|-------|-------------------------|
| 18     | 0.7837       | −0.72 | 89.0%                   |
| 19     | 0.7909       | −0.40 | 84.1%                   |
| 20     | 0.7932       | −0.30 | 77.6%                   |
| 21     | 0.7923       | −0.34 | 75.8%                   |
| 22     | 0.7970       | −0.13 | 78.1%                   |

All five  $z$ -statistics are small and negative, closely mirroring L12’s pattern (Table 4). Momentum-following performs no better, and marginally worse, than random guessing at every window tested, exactly as expected under independent draws, and consistent with our prior that this heuristic had no reason to succeed where its mirror-image opposite also failed.

**Extended window and temporal robustness check.** As with L12 (Section 7.4), we widen the lookback sweep to  $W \in \{18, \dots, 50\}$  and evaluate separately within each of W1–W5 and the full history (Figure 2).

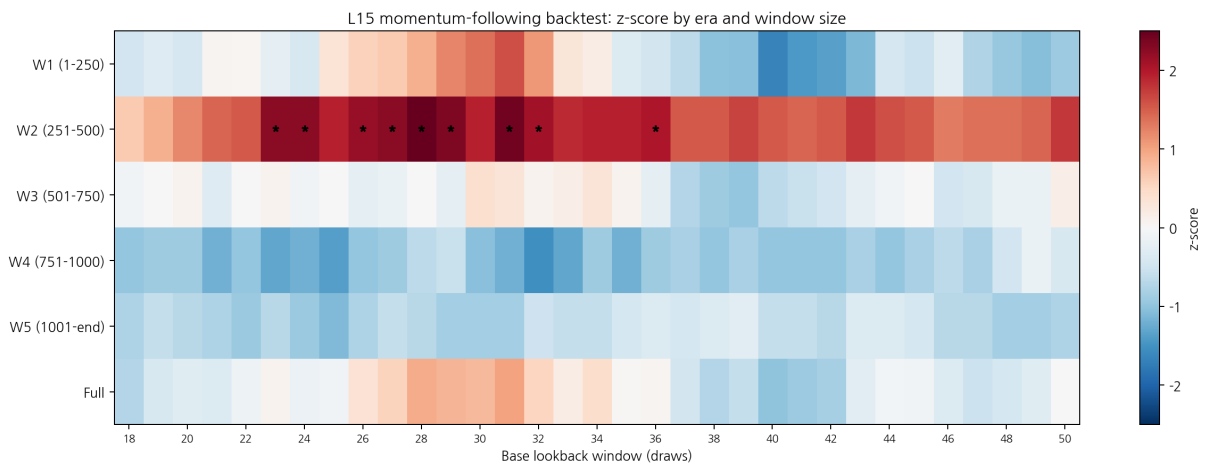


Figure 2: L15 momentum-following backtest:  $z$ -score by temporal window and lookback length, same layout as Figure 1.

| Temporal window | $n$  | mean $z$ | min $z$ | max $z$ | # sig. (of 33) |
|-----------------|------|----------|---------|---------|----------------|
| W1 (1–250)      | 232  | −0.23    | −1.67   | +1.62   | 0              |
| W2 (251–500)    | 250  | +1.73    | +0.65   | +2.50   | 9              |
| W3 (501–750)    | 250  | −0.17    | −0.97   | +0.40   | 0              |
| W4 (751–1000)   | 250  | −0.92    | −1.53   | −0.16   | 0              |
| W5 (1001–1234)  | 234  | −0.65    | −1.10   | −0.27   | 0              |
| Full (1–1234)   | 1216 | −0.09    | −1.00   | +1.02   | 0              |

Here the exception runs the other way: within W2 (draws 251–500), 9 of 33 window lengths cross  $|z| \geq 1.96$ , all *positive*, meaning momentum-following would have appeared to outperform random guessing specifically within that quarter of the history. We do not treat this as evidence for momentum any more than we treated L13’s midpoint-split correlation as evidence for enforced rebalancing (Section 8). The same logic applies: this is one era, out of five, showing a cluster of borderline significance in one direction, out of two heuristics tested in both directions (cold-number and momentum) across two independent widened sweeps. Some such clusters are expected whenever many overlapping, non-independent specifications are tested against multiple sub-periods, precisely the researcher-degrees-of-freedom risk this paper warns against elsewhere. Unlike L13, we have not subjected this specific finding to a dedicated robustness check (e.g., varying the W2 boundary itself); we flag it explicitly as an open question rather than either asserting it away or overstating it, and note that a confirmatory check in the style of L13’s split-point sweep would be a natural next step before drawing any conclusion from it.

## 8.4 L16: Within-Draw Adjacency Clustering

**Claim.** If one ball is drawn, numerically adjacent balls (e.g., 23 and 24) are more likely to also be drawn in the same round, reflecting some physical clustering in the drawing mechanism.

**Test.** For each draw, count the number of adjacent pairs  $(i, i + 1)$  both present among the six drawn balls. We derive the *exact* null distribution of this count via dynamic programming over the 45 balls in order, tracking the number of balls selected so far, whether the immediately preceding ball was selected, and the running adjacent-pair count, analogous to the exact draw-sum derivation used for L3. This gives the exact combinatorial probability mass function, not a simulated approximation. Under this exact null, the expected number of adjacent pairs per draw is  $44 \times \binom{43}{4} / \binom{45}{6} = 0.6667$ , with variance 0.5253. We test the observed mean count via a  $Z$ -test against this exact mean and variance, and separately via a binned chi-square goodness-of-fit test on the full distribution of per-draw adjacent-pair counts.

**Result.** The observed mean is 0.6613 adjacent pairs per draw against a theoretical 0.6667 ( $Z = -0.26$ ,  $p = 0.793$ ). The binned chi-square test ( $\chi^2(3) = 2.88$ ,  $p = 0.410$ , after merging the sparse  $k \geq 3$  tail) likewise shows no departure from the exact null:

| Adjacent pairs ( $k$ ) | 0     | 1     | 2     | 3+   |
|------------------------|-------|-------|-------|------|
| Observed               | 596   | 483   | 134   | 21   |
| Expected (exact)       | 581.5 | 498.5 | 138.5 | 15.6 |

We find no evidence that numerically adjacent balls co-occur more (or less) often than the exact combinatorial null predicts, refuting the adjacency-clustering claim.

## 8.5 Synthesis

None of L13–L16 survives scrutiny. L14 and L16 return unremarkable  $p$ -values on their first and only test. L13’s one nominally significant result depends entirely on an arbitrary choice of split point and disappears under a straightforward robustness check, exactly the kind of finding a careful audit is designed to catch rather than report at face value. L15’s primary specification is likewise unremarkable, though its extended window and era sweep (Section 8) surfaces one localized cluster of borderline significance (draws 251–500) that we flag as an open question pending its own robustness check rather than as a finding. Considered jointly with L1–L12, the full battery of sixteen tests, spanning static distributional checks, out-of-sample predictive backtests in both directions (cold-number and momentum) with both primary and widened-window specifications, and two structural claims about temporal and spatial clustering, converges on a single conclusion: the Korean Lotto 6/45 draw history is consistent with a fair, memoryless, independent draw mechanism, and none of the popular heuristics examined offers any exploitable structure.

## 9 Conclusion

We have presented an eleven-test statistical audit framework for lottery fairness, constructed by porting and reformulating methods from the election-forensics literature. Three methodological contributions distinguish the framework from prior approaches.

First, several tests require non-trivial adaptation to the lottery context. The corrected last-digit null (L2, L7) accounts for the structural non-uniformity of digit frequencies in the ball pool  $\{1, \dots, 45\}$ . The temporal Ljung–Box statistic (L9) is aggregated across all 45 per-ball indicator series. The structural correlation null of  $-1/44$  (L11) follows from the without-replacement sampling mechanism.

Second, the simulation-adjusted chi-square (L8), ported from [Shikano and Mack \(2011\)](#), provides a finite-sample correction to the ball-frequency test that is particularly valuable at the sample sizes (200–250 draws) typical of epoch-wise subgroup analysis.

Third, we introduce the *replication-logic criterion* as the appropriate fairness verdict in the presence of inter-test dependence: the audit result is replication-consistent with fairness if no single test fails in more than one non-overlapping temporal window. Applied to 1,234 draws of the Korean Lotto 6/45 partitioned into five windows, the criterion is satisfied: five isolated failures are observed, each in a distinct test-window cell, with no test failing more than once. This pattern is precisely what a well-calibrated audit of genuinely fair data should produce.

Fourth, we add a twelfth test of a different, predictive character: an explicit, reproducible out-of-sample backtest of the “cold-number” mean reversion heuristic widely believed by players, evaluated on the same 1,234-draw history available through July 2026. Across five lookback specifications (18–22 draws) and an explicit rule for breaking ties in the “coldest six” selection, the heuristic never outperforms, and, in every specification tested, performs slightly and non-significantly worse than, the theoretical hit rate of picking six numbers uniformly at random. This predictive falsification, obtained by methods entirely distinct from the distributional tests L1–L11, converges on the same verdict: no exploitable serial structure is detectable in the Korean Lotto 6/45 at any of the time scales examined.

Fifth, we formalize four further beliefs commonly voiced by lottery players as explicit hypothesis tests with defined null distributions (L13–L16): hot/cold reversal across time, suspiciously regular reappearance timing, momentum-following (the mirror image of L12), and within-draw adjacency clustering. Three of the four return unremarkable results outright. The fourth, L13, initially returns a nominally significant hot/cold correlation ( $p = 0.034$ ) at the exact midpoint split of the history, but this result does not survive a robustness check across other split points ( $p \approx 0.40$ – $0.50$  at the 30/70 and 70/30 splits), identifying it as a researcher-degrees-of-freedom artifact rather than genuine structure. This case illustrates, within the paper’s own results, exactly the kind of false-positive risk that motivates the replication-logic criterion of the second contribution.

The framework is implemented as an open-source Python codebase and can be applied to any 6/45-style lottery with a CSV draw history.

## A Code availability

The code is available at [https://github.com/mkimacad/mkimacad.github.io/tree/main/korean\\_lotto\\_forensics](https://github.com/mkimacad/mkimacad.github.io/tree/main/korean_lotto_forensics).

## B Exact Null Distribution of the Draw Sum

The exact probability mass function of the draw sum  $S = \sum_{j=1}^6 b_j$ , where  $\{b_1, \dots, b_6\}$  is a simple random sample without replacement from  $\{1, \dots, 45\}$ , is computed by dynamic

programming. Let  $f[j][s]$  denote the number of ways to choose  $j$  distinct elements from  $\{1, \dots, 45\}$  that sum to  $s$ . The recurrence is

$$f[j][s] = f[j-1][s-b] \quad \text{accumulated over } b = 1, \dots, 45, \quad (12)$$

with boundary condition  $f[0][0] = 1$  and  $f[0][s] = 0$  for  $s > 0$ . The probability is  $\Pr(S = s) = f[6][s] / \binom{45}{6}$ . The support of  $S$  runs from  $\sum_{i=1}^6 i = 21$  to  $\sum_{i=40}^{45} i = 255$ .

## C Structural Null Correlation for Ball Indicators

Let  $X_b = \mathbf{1}[b \in \text{draw}]$  for  $b \in \{1, \dots, 45\}$ . Then  $\mathbb{E}[X_b] = k/N = 6/45$ ,  $\text{Var}(X_b) = (k/N)(1 - k/N)$ . For  $b \neq b'$ ,

$$\text{Cov}(X_b, X_{b'}) = \mathbb{E}[X_b X_{b'}] - \mathbb{E}[X_b] \mathbb{E}[X_{b'}] = \frac{k(k-1)}{N(N-1)} - \left(\frac{k}{N}\right)^2 = -\frac{k(N-k)}{N^2(N-1)}. \quad (13)$$

Dividing by  $\text{Var}(X_b) = k(N-k)/N^2$  gives  $\rho(X_b, X_{b'}) = -1/(N-1) = -1/44$ .  $\square$

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Table 1: The sixteen audit tests, organized by conceptual domain.

| ID                                                       | Name                         | Null hypothesis tested                                                         | Source     |
|----------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------|------------|
| <i>A. Frequency domain</i>                               |                              |                                                                                |            |
| L1                                                       | Ball-frequency $\chi^2$      | Each ball appears with equal marginal probability 6/45                         | Natural    |
| L8                                                       | Simulation-adjusted $\chi^2$ | Same as L1, with finite-sample correction                                      | Ported F7d |
| <i>B. Digit-pattern domain</i>                           |                              |                                                                                |            |
| L2                                                       | Last-digit (corrected null)  | Last digits follow the non-uniform population distribution                     | Ported F2  |
| L6                                                       | 0s-and-5s bias               | Balls ending in 0 or 5 appear at the population rate 9/45                      | Ported F8  |
| L7                                                       | Last-digit mean & variance   | Last-digit mean and variance match population moments                          | Ported F9  |
| <i>C. Draw-level distribution domain</i>                 |                              |                                                                                |            |
| L3                                                       | Draw-sum distribution        | Draw sums follow the exact hypergeometric convolution                          | Natural    |
| L4                                                       | Odd/even and high/low splits | Split counts follow the hypergeometric distribution                            | Natural    |
| L5                                                       | Pairwise co-occurrence       | Each ball pair co-appears at rate $\binom{43}{4}/\binom{45}{6}$                | Natural    |
| <i>D. Dependence domain</i>                              |                              |                                                                                |            |
| L9                                                       | Temporal autocorrelation     | Per-ball indicator series are serially uncorrelated                            | Natural    |
| L11                                                      | Cross-ball correlation       | Ball indicators have structural Pearson correlation $-1/44$                    | Natural    |
| <i>E. Variance-stability domain</i>                      |                              |                                                                                |            |
| L10                                                      | Summary-statistic variance   | Variance of sum, odd count, and range match theory                             | Ported F3  |
| <i>F. Predictive (out-of-sample) domain</i>              |                              |                                                                                |            |
| L12                                                      | Cold-number mean reversion   | Out-of-sample “coldest six” picks match the next draw at the random-guess rate | New        |
| L15                                                      | Momentum-following backtest  | Out-of-sample “hottest six” picks match the next draw at the random-guess rate | New        |
| <i>G. Structural and belief-driven heuristics domain</i> |                              |                                                                                |            |
| L13                                                      | Hot/cold split correlation   | Per-ball frequencies in disjoint halves of the history are uncorrelated        | New        |
| L14                                                      | Inter-appearance gap CV      | A ball’s reappearance gaps have geometric-null variability                     | New        |
| L16                                                      | Within-draw adjacency        | Numerically adjacent balls co-occur at the exact combinatorial rate            | New        |

Table 2:  $p$ -values for all 11 tests across 5 temporal windows. **FAIL** denotes  $p$ -value  $< 0.05$ ; bold entries are failures. Tests with sub-components (L4, L7, L10) show the relevant sub-test.

| ID                                       | Test (sub-test)                 | W1          | W2   | W3          | W4          | W5          |
|------------------------------------------|---------------------------------|-------------|------|-------------|-------------|-------------|
| <i>A. Frequency domain</i>               |                                 |             |      |             |             |             |
| L1                                       | Ball frequency $\chi^2(44)$     | .980        | .847 | .961        | .719        | .631        |
| L8                                       | Sim.-adj. $\chi^2$              | .934        | .678 | .910        | .492        | .432        |
| <i>B. Digit-pattern domain</i>           |                                 |             |      |             |             |             |
| L2                                       | Last digit $\chi^2(9)$          | .814        | .357 | .383        | .486        | .875        |
| L6                                       | 0s-and-5s $Z$                   | .651        | .071 | .606        | <b>.024</b> | .883        |
| L7                                       | LD mean $Z$                     | .691        | .553 | .853        | .697        | .267        |
| L7                                       | LD variance $\chi^2$            | .610        | .560 | .260        | .915        | .954        |
| <i>C. Draw-level distribution domain</i> |                                 |             |      |             |             |             |
| L3                                       | Draw sum $\chi^2$               | .473        | .677 | .651        | .119        | .917        |
| L4                                       | Odd/even split $\chi^2$         | .634        | .874 | .734        | .471        | <b>.033</b> |
| L4                                       | High/low split $\chi^2$         | .099        | .834 | .782        | .598        | .789        |
| L5                                       | Pairwise co-occurrence $\chi^2$ | .957        | .848 | .996        | .894        | .572        |
| <i>D. Dependence domain</i>              |                                 |             |      |             |             |             |
| L9                                       | Temporal autocorr. (Ljung–Box)  | .633        | .308 | <b>.045</b> | .880        | .851        |
| L11                                      | Cross-ball correlation $\chi^2$ | .997        | .996 | 1.000       | 1.000       | .999        |
| <i>E. Variance-stability domain</i>      |                                 |             |      |             |             |             |
| L10                                      | Variance (draw sum)             | <b>.014</b> | .858 | .641        | .873        | .905        |
| L10                                      | Variance (odd count)            | .346        | .856 | .465        | .138        | .742        |
| L10                                      | Variance (draw range)           | .126        | .999 | .390        | .454        | <b>.015</b> |
| <i>Failures per window</i>               |                                 | 1           | 0    | 1           | 1           | 2           |

Table 3: Failure map across tests and windows. A filled circle ( $\bullet$ ) indicates  $p$ -value  $< 0.05$ ; an open circle ( $\circ$ ) indicates  $p$ -value  $< 0.10$ . No test fails in more than one window.

| Test (sub-test)  | W1        | W2      | W3        | W4        | W5        |
|------------------|-----------|---------|-----------|-----------|-----------|
| L4 - Odd/even    |           |         |           |           | $\bullet$ |
| L6 - 0s-and-5s   |           | $\circ$ |           | $\bullet$ |           |
| L9 - Autocorr.   |           |         | $\bullet$ |           |           |
| L10 - Sum var.   | $\bullet$ |         |           |           |           |
| L10 - Range var. |           |         |           |           | $\bullet$ |

Table 4: Out-of-sample backtest of the cold-number mean-reversion heuristic.  $z$  compares the observed mean match count to the theoretical random-guess baseline of 0.8000 matches, using the exact hypergeometric variance and treating each forecast as an independent trial; because successive forecasts share heavily overlapping lookback windows, these  $z$ -statistics understate the true sampling uncertainty and should be read as indicative rather than exact.

| Window $W$ | $N$  | Mean matches | $z$ vs. $H_0$ | Avg. extra draws | % needing ext. |
|------------|------|--------------|---------------|------------------|----------------|
| 18         | 1216 | 0.7566       | -1.93         | 8.87             | 92.4%          |
| 19         | 1215 | 0.7621       | -1.68         | 8.32             | 93.6%          |
| 20         | 1214 | 0.7611       | -1.73         | 7.64             | 94.4%          |
| 21         | 1213 | 0.7659       | -1.52         | 6.97             | 92.3%          |
| 22         | 1212 | 0.7690       | -1.38         | 6.56             | 89.6%          |

Table 5: Match-count distribution (% of forecasts) by window, versus the exact theoretical distribution under the null of no predictive signal.

| Matches ( $k$ ) | $W=18$ | $W=19$ | $W=20$ | $W=21$ | $W=22$ | Theory ( $H_0$ ) |
|-----------------|--------|--------|--------|--------|--------|------------------|
| 0               | 42.93  | 42.47  | 42.67  | 42.13  | 41.91  | 40.06            |
| 1               | 40.46  | 40.82  | 40.53  | 41.06  | 41.17  | 42.41            |
| 2               | 14.80  | 14.90  | 14.99  | 15.09  | 15.18  | 15.15            |
| 3               | 1.64   | 1.65   | 1.65   | 1.57   | 1.57   | 2.24             |
| 4               | 0.16   | 0.16   | 0.16   | 0.16   | 0.17   | 0.14             |
| 5               | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | $\approx 0.00$   |
| 6               | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | $\approx 0.00$   |